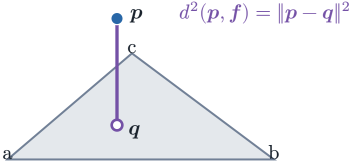
A. Exact point–triangle primitive

the closest point may lie on the face, an edge, or a vertex



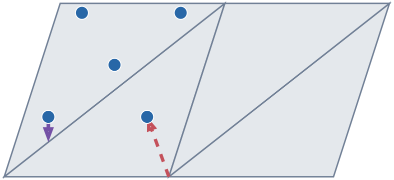
The backend evaluates squared Euclidean distance to the triangle itself, not distance to its supporting plane or centroid.

B. Two directional reductions

the arrows reverse which population contributes summands

solid: $p \rightarrow$ nearest triangle

dashed: triangle $\rightarrow$ nearest p



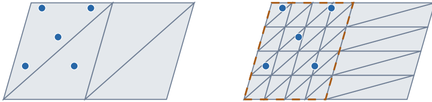
One exact witness is shown per direction; the implementation reduces over every point or every face.

C. Controlled tessellation check

same planar surface + same points; only tessellation changes

4 equal-area **40 non-uniform**

left region: 32/40 equal-weight faces



$D_{M \rightarrow P}$ =0.03640 m^2 $D_{M \rightarrow P}$ =0.02284 m^2

The left region contributes 32 of 40 equally weighted face summands.

Controlled conclusion. $D_{P \rightarrow M}$ =0.00640 m^2 in both; $D_{M \rightarrow P}$: 0.03640 $\rightarrow$ 0.02284 m^2 after retessellation.